Regulation of Chemical Transformation in Designer Peptide Biomolecular Condensates

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Abstract

Biomolecular condensates, formed through liquid-liquid phase separation (LLPS), serve as dynamic platforms for biochemical regulation. Inspired by these natural systems, we developed designer peptide-based condensates to modulate chemical transformations, focusing on the

Cu(I)-catalyzed azide-alkyne cycloaddition click reaction between hydrophobic reactants as a model system. By incorporating varying number of isoleucine residues into peptide sequences, we tuned hydrophobicity of the condensates, influencing reaction rate and conversion. Condensates formed by a peptide with a single isoleucine enhanced reactant recruitment and reaction efficiency, while excessive hydrophobicity resulted in solid-like condensates that impeded catalysis. Notably, we deciphered the factors that affect reactant recruitment, eventually resulting in a complete spatial regulation of product localization inside the condensates. The study highlights the critical balance of hydrophobicity for optimal reaction performance in condensates, demonstrating a new approach to regulate chemical transformations in water-based systems. This work establishes a foundation for engineering biomolecular condensates as green chemistry platforms and reaction vessels for applications in biotechnology and biomedicine.

Introduction

The compartmentalization of biochemical reactions within spatially distinct cellular environments is a hallmark of life, allowing for precise spatiotemporal regulation of complex processes[1]. For example, ATP synthesis occurs within mitochondria, leveraging the proton gradient to drive energy production, while proteolysis is compartmentalized in lysosomes, isolating degradative enzymes to ensure cellular homeostasis. Beyond membrane-bound organelles, many cellular functions are orchestrated within membraneless organelles, or biomolecular condensates[2], which are formed through liquid-liquid phase separation (LLPS) and provide dynamic, compartmentalized environments for biochemical transformations. Such condensates, including nucleoli and nuclear stress bodies, create specialized

microenvironments for processes like ribosome assembly[3] and phosphorylation of splicing factors[4], respectively.

The physical basis of LLPS lies in networks of weak, non-covalent interactions among intrinsically disordered proteins (IDPs) and nucleic acids[2], including electrostatic interactions, hydrogen bonds, and π–π stacking[5,6]. This interplay of interactions generates dense, liquid-like phases with unique physicochemical properties[7], which enable the selective recruitment, sequestration, and transformation of biomolecules[8]. These features have inspired the design of synthetic biomolecular condensates for diverse applications, including as enzyme microreactors, molecular delivery systems[9], and platforms for catalytic transformations[10] including compartmentalization of organic reactions such as aldol and hydrazone formation[11], redox reactions[12,13], imine synthesis[14], light-triggered radical polymerization[15], retro-Diels−Alder reaction[16], and selective amide bond formation[17].

Despite the potential of synthetic condensates, understanding the determinants that govern their performance as reaction vessels—particularly how dense-phase chemical properties affect reactant recruitment, reaction efficiency, and product localization—remains limited. Of particular interest is the role of hydrophobic microenvironments, which can recruit hydrophobic reactants and mediate reactions in aqueous systems, avoiding the need for organic solvents. This approach aligns with principles of green chemistry by reducing environmental impact while enhancing reaction efficiency.

Here, we present a systematic exploration of hydrophobicity as a design parameter in synthetic peptide-based condensates. Employing chemically diverse building blocks such as peptides[18,19] enables the engineering of condensates with a rich set of chemical properties[20–22]. Using minimalistic peptides, we engineered condensates with tunable hydrophobicity by incorporating varying numbers of isoleucine residues into the peptide sequence. As a model

system, we employed the Cu(I)-catalyzed azide-alkyne cycloaddition i.e. click chemistry reaction[23], a widely used transformation in drug discovery, proteomics, and materials science[24,25]. This reaction, between two hydrophobic azide-alkyne reactants, forms a traceable aromatic triazole product under mild conditions in aqueous environments, provides an ideal platform for studying the interplay of condensate microenvironment and reaction kinetics.

Our findings demonstrate that condensates with moderate hydrophobicity acheive an optimal balance between reactant recruitment and catalytic efficiency. In contrast, excessive hydrophobicity leads to the formation of solid-like condensates that hinder dynamic exchanges, while overly hydrophilic systems are ineffective at enhancing reactant encapsulation. These findings highlight the critical balance of hydrophobicity for optimal reaction performance in condensates, setting the ground for the rational design of biomolecular condensates as reaction vessels, to enable precise control over reaction localization, rate, and conversion. By leveraging peptide-based condensates as versatile and sustainable platforms, this work underlines their potential as green chemistry solutions for catalysis in aqueous systems.

Results

Design of peptide condensates with varying hydrophobicity

We aimed to develop biomolecular condensates that function as reaction vessels and facilitate the spatial modulation of chemical transformations. Building on our prior work, we utilized a minimalistic 14-mer peptide designed to promote LLPS via simple coacervation, forming biomolecular condensates[20]. The peptide sequence comprises three repeats of glycine-arginine (Gly-Arg) dyads, which are common in low-complexity domains of IDPs and promote phase separation by providing backbone flexibility. Additionally, the peptide includes three aromatic amino acids–two tryptophans (Trp) and one tyrosine (Tyr)–which drive π-π interactions among

themselves and with the Arg side chains[20]. To further enhance LLPS, the peptide incorporates an elastin-like polypeptide domain (Pro-Gly-Val-Gly), known to promote phase separation through simple coacervation[26,27]. To investigate how condensate properties, particularly hydrophobicity, influence reaction conversion, we designed a small library of peptide building blocks with varying hydrophobicity by introducing different numbers of the aliphatic amino acid isoleucine (Ile). The primary peptide, lacking Ile, was named WGI-0. Modified peptides with one or two Ile residues were designated WGI-1 and WGI-2, respectively (Fig. 1a, Fig. S1).

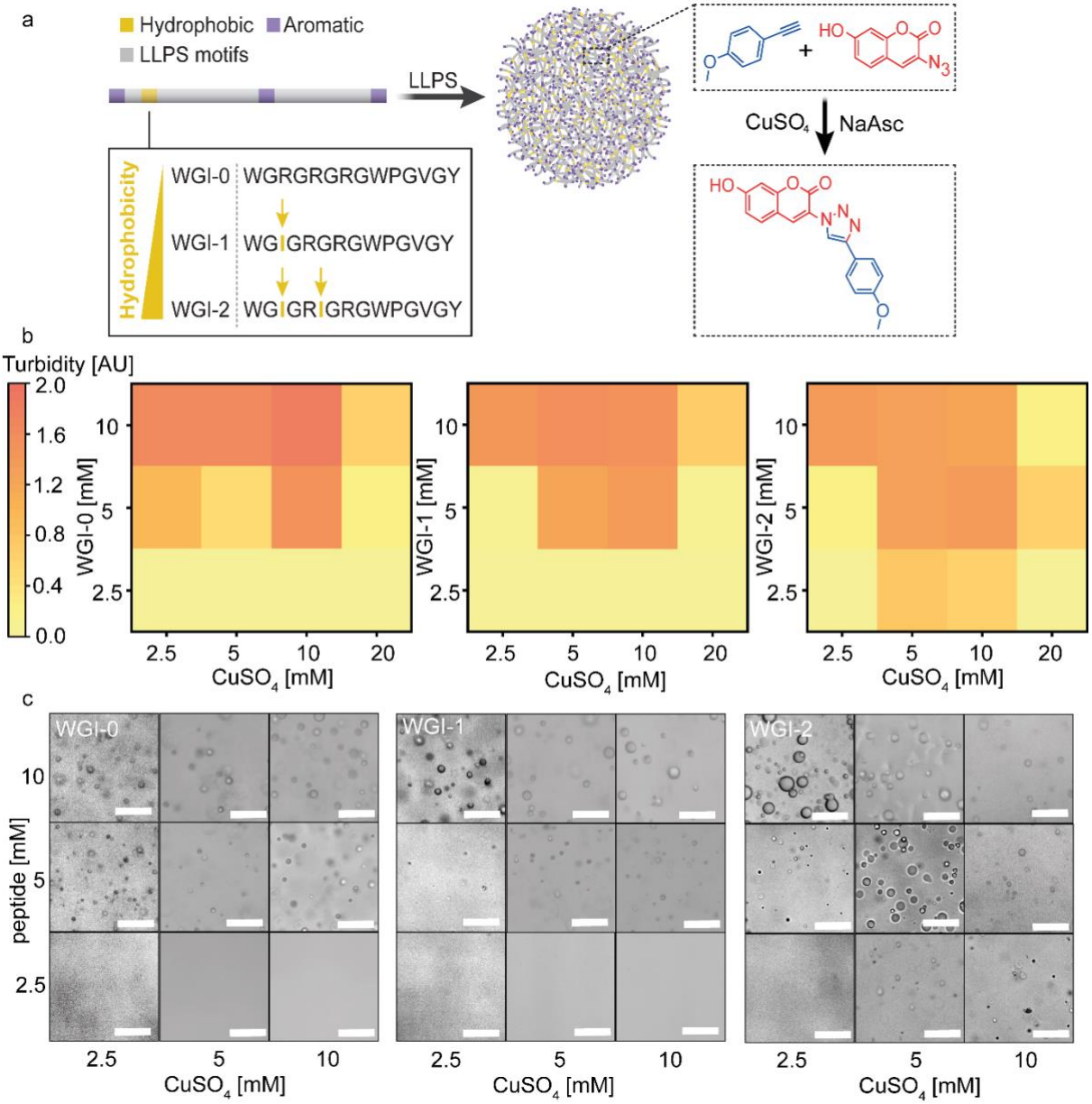

Figure 1. Hydrophobicity of peptide condensates regulates chemical reactions. a. Schematics of the suggested system: LLPS-promoting peptide building blocks with varying level of hydrophobicity form biomolecular condensates for regulation of a click chemistry reaction. **b.** Phase diagram heatmaps of WGI-0, WGI-1, and WGI-2 as a function of peptide and $CuSO_4$ concentration. Peptides were dissolved in 20 mM citrate buffer pH 6 (final pH is 5.5). Turbidity intensity was monitored at λ=400 nm. **c**. Microscopy analysis of the peptides as a function of peptide and $CuSO_4$ concentration. Scale bar=25 μm.

As a model reaction system, we focused on the Cu(I)-catalyzed reaction between azide and terminal alkyne to form a 1,2,3-triazole (Fig. 1a, Fig. S2). Click reactions are well characterized and widely used in drug discovery and proteomic studies[23], making them an ideal model for studying compartmentalized drug synthesis. Since the reaction requires Cu(I) as a catalyst, we employed $CuSO_4$,[28] which acts both as a catalyst and an initiator for LLPS by screening the charges of the cationic peptides, thereby promoting intermolecular interactions and simple coacervation. We first evaluated the LLPS propensity of each peptide as a function of peptide and $CuSO_4$ concentration at pH 5.5. Peptides were dissolved in citrate buffer at pH 6, which decreases to 5.5 after dissolution, at final concentrations of 2.5 mM, 5 mM or 10 mM. Phase diagrams were constructed by measuring sample turbidity (λ=400 nm), following addition of $CuSO_4$, as light scattering by condensates corresponds to turbidity (Fig. 1b). Microscopy analysis confirmed that the turbidity resulted from condensate formation rather than aggregate formation (Fig. 1c). For WGI-0 and WGI-1, the addition of $CuSO_4$ triggers phase separation and condensate formation (Fig. S3a-b). In contrast, WGI-2, above 5 mM, undergoes LLPS immediately when dissolved in buffer (Fig. S3c). This indicates that the higher hydrophobicity of WGI-2 promotes attractive forces which overcome the electrostatic repulsion between the peptide molecules, resulting in LLPS and condensate formation.

These turbidity and microscopy analyses revealed that the LLPS propensity of all three peptides was concentration dependent, with turbidity increasing as a function of peptide concentration. Notably, hydrophobicity played a key role, as evidenced by differences in phase diagram boundaries and turbidity intensity between peptides. For instance, condensates formed by WGI-1/CuSO4 (5 mM/5 mM) exhibited 2.4-fold higher turbidity compared to those formed by WGI-0 under the same conditions (Fig. 1b). Increasing the number of Ile residues from one (WGI-1) to two (WGI-2) further shifted the phase diagram, enabling LLPS at a lower peptide concentration of 2.5 mM for WGI-2 (Fig. 1b, c). These results establish a clear correlation between peptide hydrophobicity and LLPS propensity.

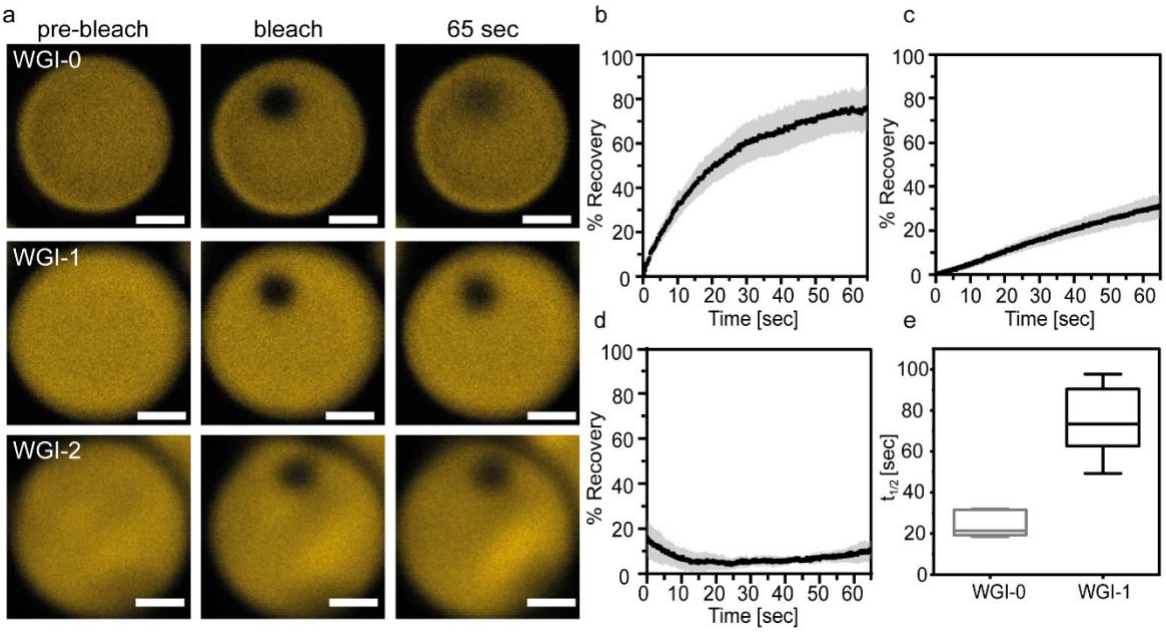

Figure 2. Condensate dynamics is directly affected by the number of Ile in the peptide sequence. FRAP analysis of WGI-0, WGI-1, and WGI-2 condensates using rhodamine B (10 μM) as a fluorescent payload. Condensates were formed by 10 mM of each peptide and 10 mM $CuSO_4$ **a**. LSCM images of condensates before, immediately after, and 65 sec after photobleaching of rhodamine B. Scale bars=2 μm. **b-d**. Recovery plots of (**b**) WGI-0, (**c**) WGI-1, and (**d**) WGI-2 condensates. **e**. $t_{1/2}$ values of WGI-0 and WGI-1 condensates. Recovery plots

and $t_{1/2}$ values represent average of n=5 condensates for WGI-0, n=9 for WGI-1, n=8 for WGI-2.

The diffusion in the condensates correlates with peptide hydrophobicity

To analyze the material properties of the condensates, we performed fluorescence recovery after photobleaching (FRAP) analysis using confocal laser scanning microscopy (CLSM). Fluorescence of rhodamine B in the condensates was monitored before and after photobleaching to generate recovery plots. Condensates were formed by 10 mM peptide concentration, as at 5 mM, WGI-0 and WG-1 condensates were either too small or too dynamic for proper analysis. The results demonstrate that peptide hydrophobicity significantly impacts condensate dynamics (Fig. 2a-c). Increasing the number of Ile residues in the peptide building block from zero to one increased the $t_{1/2}$ by 4.2-fold, from 20.5±5.5 s for WGI-0 to 87.3±18.6 s for WGI-1 (Fig. 2e). This trend was further supported by a twofold higher total recovery in WGI-0 condensates (79.5±8.3%) compared to WGI-1 condensates (38.5±6.7%). Further increasing the number of Ile residues to two drastically altered the condensate properties, as WGI-2 condensates exhibited no fluorescence recovery after photobleaching (Fig. 2d). These results suggest that WGI-2 forms viscous dense phases rather than dynamic liquid assemblies.

Phase separating peptide systems enhance reaction conversion

To analyze the impact of phase separated systems on reaction conversion and rate, we utilized the Cu(I)-catalyzed click reaction between 3-azido-7-hydroxycoumarin (AHC) and 4-ethynylanisole to form the fluorescent triazole product 7-hydroxy-3-(4-(4-methoxyphenyl)-1H-1,2,3-triazol-1-yl)-2H-chromen-2-one (λ_{ex}=330 nm, λ_{em}=460, Fig. S4, Fig. S5). The reaction involves the reduction of Cu(II) to Cu(I) by sodium ascorbate (NaAsc), which activates the ethynyl group, allowing azide to form the triazole ring. To prevent oxidation of Cu(I) to Cu(II) reactions were conducted under an argon (Ar) atmosphere. Reactants were used

at final concentration of 0.5 mM, as higher concentrations destabilize the condensates during the reaction (Fig. S6).

Peptides were dissolved in citrate buffer at a final concentration of 5 mM with 10 $CuSO_4$, followed by the addition of reactants and NaAsc. Reactants were dissolved in *tert*-butanol (t-BtOH), where all reactions contained a final concentration of 5% of the solvent. The peptide solutions were incubated with NaAsc under Ar for 30 minutes before the addition of $CuSO_4$ at a final concentration of 10 mM (Fig. S7). Reaction kinetics were monitored using HPLC by measuring product formation over time. Reactions were quenched at specific intervals with acetonitrile (ACN) before injection, thus this analysis represents product formation in the heterogenous mixture of both the dilute and dense phases. Conversion was calculated using a calibration curve of the product (Fig. S8a). The reaction in citrate buffer and in WGI-2 condensate system showed the lowest conversion (28±4.2% and 16±3%, respectively). Higher conversions of 32±5.2% and 38±7% were observed in ddw with 5% t-BtOH and in the WGI-0 phase separated system, respectively. Notably, the WGI-1 condensate system achieved the highest conversion of 59.6±8.9%, although the reaction rate remained comparable to WGI-0 and ddw systems (Fig. 3a, b).

Brightfield microscopy confirmed that WGI-0 and WGI-1 condensates remained stable throughout the reaction, while WGI-2 condensates aggregated over time (Fig. 3c). Further analysis revealed that WGI-2 aggregation was product-induced as no such aggregation was observed with NaAsc or the reactants alone (Fig. S9), potentially explaining its lower conversion. When reactions were conducted below the saturation concentration (C_{sat}), where no condensates form, the conversion rates for WGI-0 and WGI-1 peptides were similar to those in buffer, highlighting the role of the dense phase in enhancing reaction efficiency (Fig. 3d, e).

The lower reaction conversion in citrate buffer suggests that citric acid ions inhibit the reaction, possibly by acting as ligands to Cu(II)/Cu(I) and competing with its interaction with the ethynyl group.

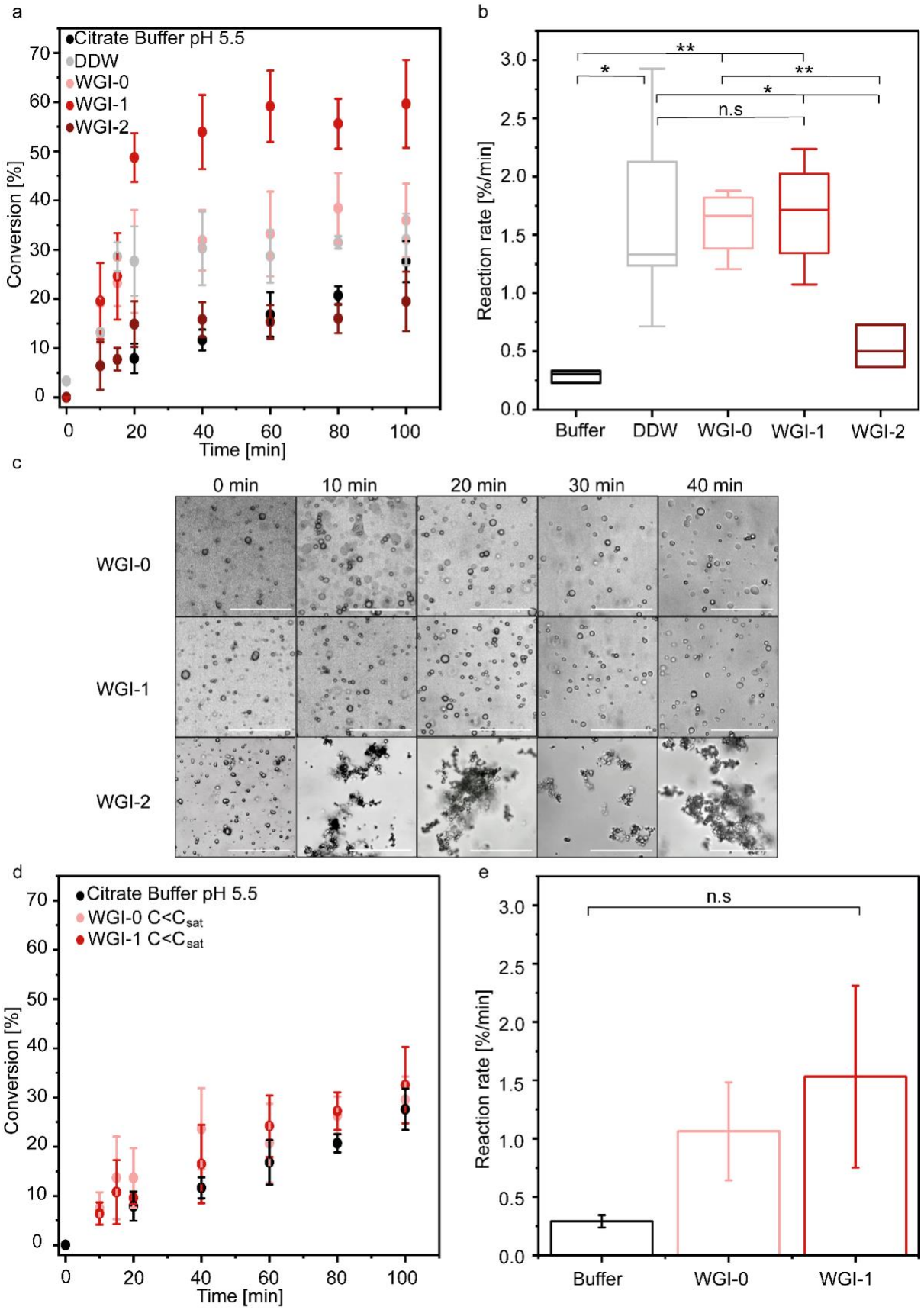
a
Citrate Buffer pH 5.5
DDW
WGI-0
WGI-1
WGI-2
Conversion [%]
Time [min]
b
Reaction rate [%/min]
Buffer
DDW
WGI-0
WGI-1
WGI-2
n.s
c
0 min
10 min
20 min
30 min
40 min
WGI-0
WGI-1
WGI-2
d
Citrate Buffer pH 5.5
WGI-0 C<Csat
WGI-1 C<Csat
Conversion [%]
Time [min]
e
Reaction rate [%/min]
Buffer
WGI-0
WGI-1
n.s

Figure 3. Reaction conversion kinetics in phase separated systems. a-b. Reaction conversion over time (**a**) and reaction initial rate (**b**) in either citrate buffer, ddw, or in phase separated systems formed by 5 mM of WGI-0, WGI-1, of WGI-2, formed in citrate buffer pH 5.5. All reactions contain 5% of t-BtOH. Analyses were performed by HPLC, following quenching of the reaction at each time point. Values represent averages (a) or box plot (b) of either three, four, or 5 independent analyses for the reaction in buffer, each of the LLPS system, and ddw, respectively. Error bars in panels a and d represent S.D. **c**. Brightfield microscopy analysis of condensates over reaction time. Scale bars=25 μm. **d-e**. Reactions in the presence of WGI-0 or WGI-1 at a concentration below C_{sat} (2mM), where no condensates form. Values represent averages of three independent analyses, error bars represent S.D. $*p$ value <0.05, $**p$ value <0.01.

Since citrate buffer facilitates phase separation through charge screening[20], it is plausible that in the WGI-0 and WGI-1 systems, citric acid ions interact preferentially within the dense phase rather than binding to Cu(II)/Cu(I). This interaction may mitigate the inhibitory effect of citric acid ions, resulting in higher conversions in these systems. However, the significantly higher conversion observed in the WGI-1 system compared to ddw suggests that the dense phase of this peptide provides additional functionality beyond merely sequestering citric acid ions.

Regulation of reactant recruitment to the condensates

To investigate the mechanism by which WGI-1 increases reaction conversion, we analyzed the encapsulation efficiency (EE) of AHC in WGI-0 and WGI-1 condensates. In citrate buffer (pH 5.5), the EE for condensates formed by 5 mM WGI-0 was very low (7±5.4%), increasing to 26±15.5% with 10 mM WGI-0 (Fig. 4d). Suspecting that divalent Cu(II) ions might limit EE,

we tested condensates formed with monovalent $NaSO_4$ instead of $CuSO_4$, but this resulted the same EE (8±9%). Replacing SO_4^{2-} with $(Cl^-)_2$ using $CuCl_2$ led to an EE of 18±9%, indicating that ions in solution do not limit reactant recruitment (Fig. 4a). To determine whether electrostatic repulsion of the reactant from the cationic peptide was a factor, we compared the EE of AHC with that of coumarin. Surprisingly, coumarin showed even lower EE than AHC (Fig. 4b), suggesting the azido group does not hinder recruitment. Adjusting the pH to neutral did not improve EE either (Fig. 4c). However, increasing the peptide concentration to 20 mM raised EE to 43±6.7% (Fig. 4d).

Considering the inhibitory effect of citrate buffer on reaction conversion (Fig. 3a), we next analyzed EE in 2-(N-morpholino)ethanesulfonic acid (MES) buffer, which does not act as a Cu(II)/Cu(I) ligand. At acidic pH, no improvement in EE was observed even at 20 mM WGI-0 (Fig. 4e). However, at pH 7.5, EE increased to 64±2.6% in 20 mM WGI-0 condensates (Fig. 4f).

We then assessed whether the enhanced reaction conversion in WGI-1 condensates correlated with higher EE. In citrate buffer at acidic pH, the EE for 5 mM WGI-1 was 35.8±3.0%, but increased to 65.0±7.5% with 20 mM WGI-1 (Fig. 4g). Using MES buffer, acidic pH resulted in consistent EE (44±3.6%, 45.6±3.7% and 48±.79% for 5, 10, 20 mM respectively) across all concentrations (Fig. 4h), while increasing the pH to 7.5 yielded EE up to 91.6±1.6% for 20 mM WGI-1 (Fig. 4i).

The low EE (7.0±5.4%) in 5 mM WGI-0 condensates formed in citrate buffer suggests that the reaction occurs entirely in the dilute phase (Fig. 4d). Conversely, the 36% EE in 5 mM WGI-1 condensates implies reactions occur in both dilute and dense phases, contributing to higher conversion (Fig. 3a). Notably, reaction rates in phase-separating systems were comparable to those in ddw (Fig. 3b). However, WGI-1 condensates achieved almost twofold higher

conversion at t=20 min (Fig. 3a), which might be explained by slow diffusion (Fig. 2c, e) of the accumulating product from the dense to the dilute phase and thereby increasing the conversion in the dense phase. These findings suggest that dense-phase properties enhance reaction efficiency and conversion.

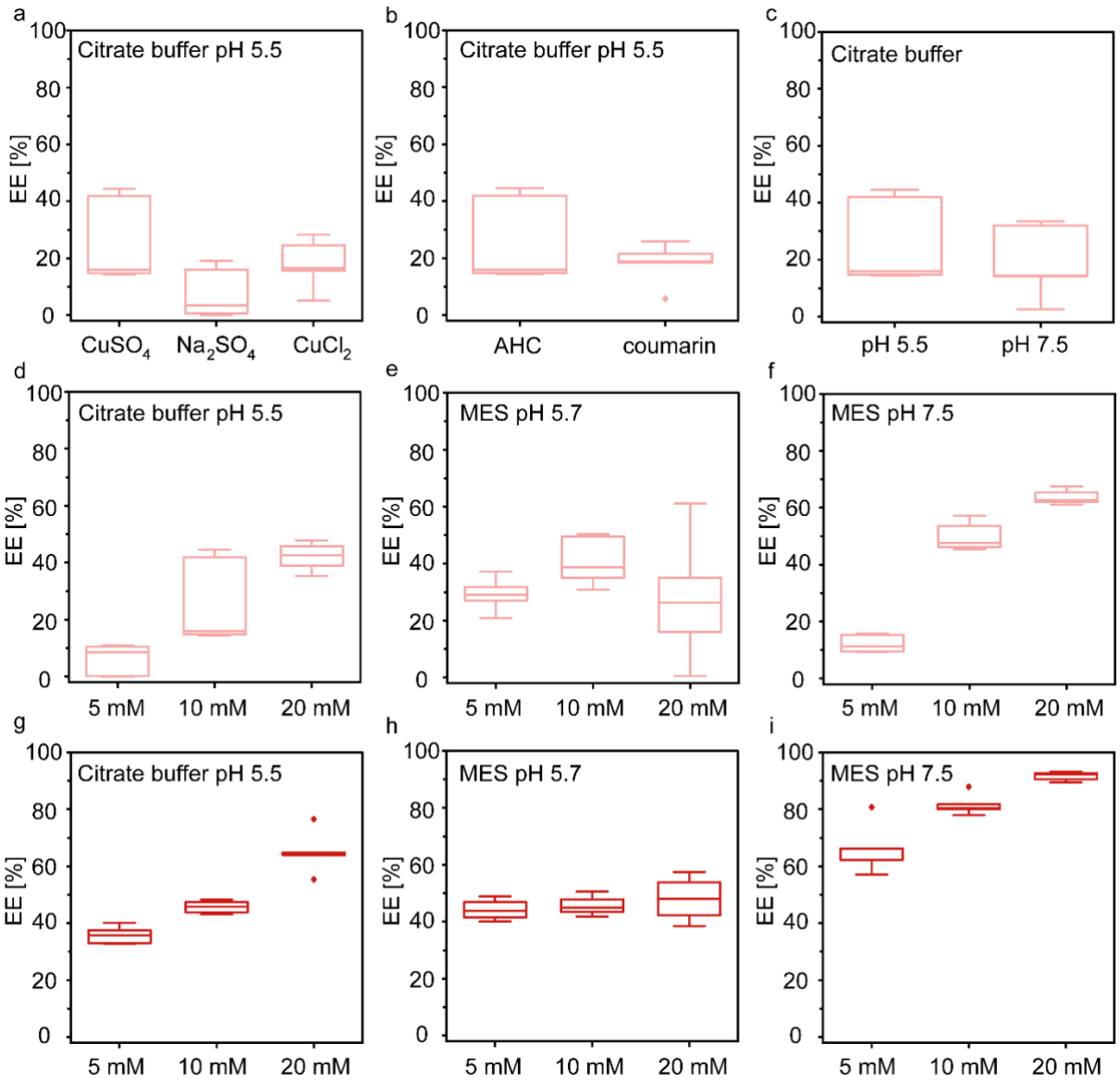

Figure 4. Deciphering the factors affecting reactant recruitment to condensates. Encapsulation efficiency (EE) of AHC in (**a-f**) WGI-0 and (**g-i**) WGI-1 condensates. **a-f.** The effect of salts (**a**), the azide group of the reactant (**b**), pH (**c**), peptide concentration (**d-f**), and

buffer composition (**e, f**) were analyzed in WGI-0 condensates. **g-i.** The effect of peptide concentration and buffer composition (citrate vs. MES buffer) was analyzed in WGI-1 condensates. Values presented as box plots based on n=4-6. Brightfield images of the different conditions are shown in Fig. S10.

Reactions in phase separating systems with high reactant recruitment

The high recruitment efficiencies of WGI-0 (EE = 64±2.6%) and WGI-1 (EE = 91.6±1.6%) at 20 mM in MES buffer (pH 7.5) significantly enhanced reaction performance. Reactions conducted within WGI-0 and WGI-1 LLPS systems achieved conversion rates of 43.9±10.2% and 47.5±6.0%, respectively, with reaction rates increased by 3.1 and 2.8-fold respectively compared to those in MES buffer alone (25% conversion, Fig. 5a). Notably, the reaction rates in both condensates were similar, 3.3±1 and 3.7±1 %/min respectively (Fig. 5b), and were substantially higher than those observed in citrate buffer (Fig. 3b).

Brightfield microscopy further revealed that the condensates exhibited a decrease in diameter over time, likely due to coalescence after approximately 10 minutes (Fig. 5c). Although the overall conversions in high-recruitment LLPS systems were comparable to those observed in citrate buffer (Fig. 3a), the twofold faster reaction rates highlight the critical role of high reactant recruitment in accelerating reaction kinetics.

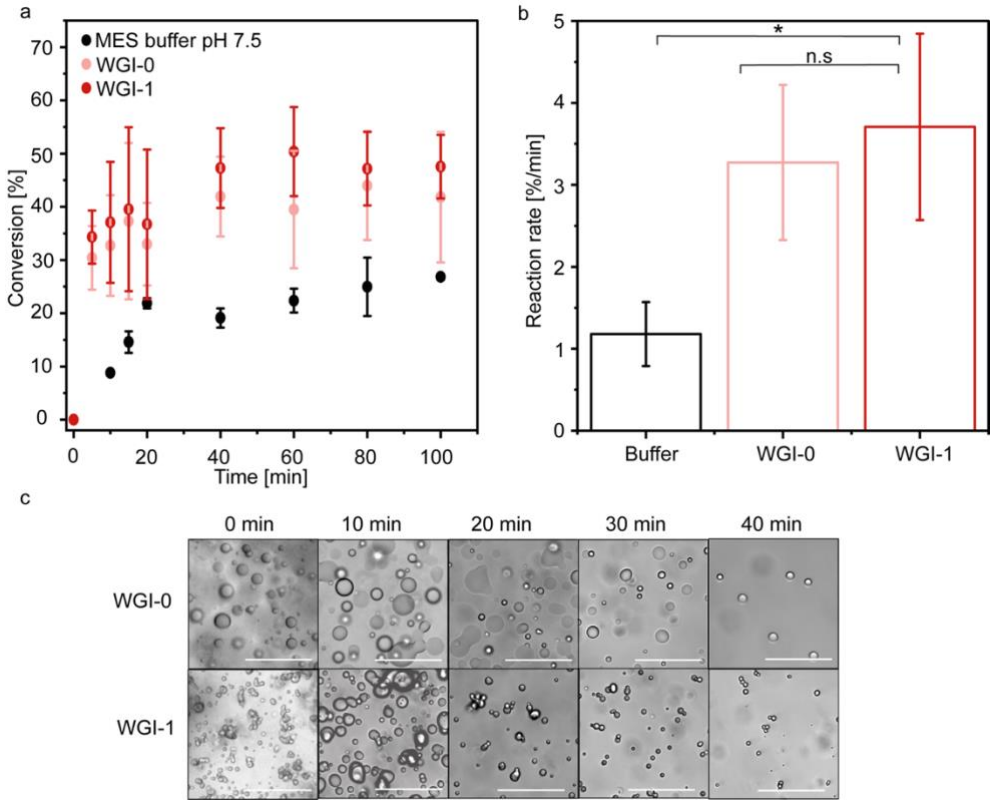

Figure 5. Reaction conversion kinetics in phase separated systems. Reaction conversion over time (**a**) and reaction rate (**b**) in either MES buffer at pH 7.5 or in phase separated systems formed by 20 mM of WGI-0, WGI-1, formed in MES buffer pH 7.5. All reactions contain 5% of t-BtOH. Analyses were performed by HPLC, following quenching of the reaction at each time point. Values represent averages of three independent analyses, error bars represent S.D. **c**. Brightfield microscopy analysis of condensates over reaction time in MES buffer pH 7.5. Scale bars=25 µm. $*p$ value <0.05.

Regulation of product localization by reactant recruitment and ion composition in LLPS systems

To investigate the localization of the fluorescent product over reaction time in LLPS systems, CLSM analysis was performed in citrate and MES buffers. In citrate buffer, product fluorescence was observed in both the dilute and dense phases, with significant fluorescence detected in the dilute phase of both WGI-0 and WGI-1 LLPS systems at t=10 min (Fig. 6a).

In contrast, product fluorescence was predominantly localized in the dense phase of both WGI-0 and WGI-1 condensates in MES buffer, with an 11-fold increase in fluorescence intensity in the dense phase compared to the dilute phase for WGI-1 (Fig. 6b). Notably, the condensate diameters of both WGI-0 and WGI-1 decreased over reaction time in MES buffer, likely due to rapid coalescence, which may explain the observed decrease in fluorescence intensity over time.

The high fluorescence signal detected at t=0 min in WGI-1 condensates in MES buffer corresponds with the high conversion observed in the heterogeneous LLPS mixture of WGI-1 at the same time point (Fig. 5a). Furthermore, the relative fluorescence in the dense phase of both condensate systems in citrate buffer was significantly lower compared to MES buffer. Among all conditions, the highest relative fluorescence in the dense phase was observed in WGI-1 condensates in MES buffer (Fig. 6c).

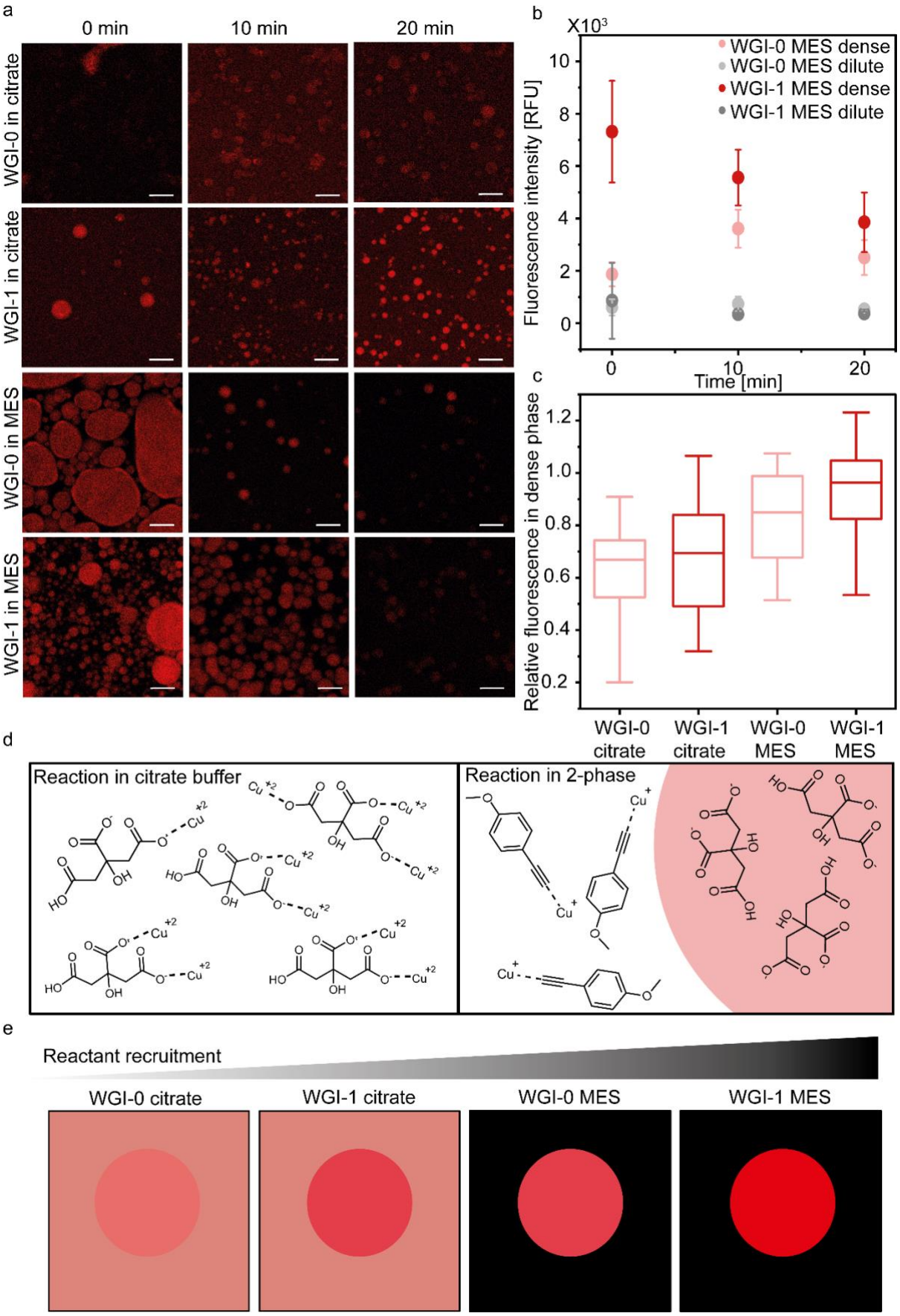
a
0 min
10 min
20 min
WGI-0 in citrate
WGI-1 in citrate
WGI-0 in MES
WGI-1 in MES
b
X10³
WGI-0 MES dense
WGI-0 MES dilute
WGI-1 MES dense
WGI-1 MES dilute
Fluorescence intensity [RFU]
Time [min]
c
Relative fluorescence in dense phase
WGI-0 citrate
WGI-1 citrate
WGI-0 MES
WGI-1 MES
d
Reaction in citrate buffer
Reaction in 2-phase
e
Reactant recruitment
WGI-0 citrate
WGI-1 citrate
WGI-0 MES
WGI-1 MES

Figure 6. Product localization is regulated by ion composition and reactant recruitment. CLSM analysis of product fluorescence in WGI-0 and WGI-1 LLPS systems in either citrate buffer (pH=5.7) or MES buffer (pH=7.5). Condensates were formed in citrate and MES buffer by 5 mM and 20 mM of each peptide, respectively. **a.** CLSM images collected λex=405 nm. Scale bars=10 μm. **b.** Fluorescence intensity of product in the dense and dilute phase of WGI-0 and WGI-1 in MES buffer (pH=7.5), calculated based on CLSM analysis. Values represent averages of n=20-53 droplets, error bars represent S.D. **c.** Relative fluorescence intensity of the product in the dense phase/total fluorescence intensity of dense and dilute phases of WGI-0 and WGI-1 in citrate or MES buffers. Values represent box plots based on n=20-53 droplets.

Discussion

In this work, we show that the distinctive chemical environments in designed biomolecular condensates affect reaction rate and conversion. The tunable hydrophobicity in the condensed phase is achieved by varying the number of the aliphatic amino acid Ile in the peptide sequence. To create biomolecular condensates that will provide hydrophobic microenvironment for compartmentalization of click reaction, we utilized cationic and aromatic LLPS-promoting peptides, where the $CuSO_4$ salt used in this system serves a dual function, as the initiator of LLPS by charge screening of the cationic peptide molecules and as the catalyst of the reaction in the Cu(I) form. The results from the phase diagrams and FRAP analyses show that incorporation of a single Ile to the peptide sequence does not significantly change the LLPS propensity, while incorporation of two Ile decreases the saturation concentration (C_{sat}) and increases the durability of the condensates. Moreover, incorporation of two Ile significantly changes the material properties of the condensates, from dynamic droplets (WGI-0, WGI-1) to solid-like assemblies (WGI-2).

The reaction kinetics results show that ion composition affect reaction efficiency, where citric acid ions restrict the click reaction likely by binding to Cu(I) ions. This restriction does not exist in the two-phase systems in which the affinity of citric acid ions to the ionizable side chains of the peptides compete with the affinity to Cu(I) (Fig. 6d). An additional possible explanation to the enhanced efficiency of the reaction with WGI-1 might be the hydrophobic effect[29], as was suggested for Diels-Alder reactions in water[30]. This effect is driven by entropy increase due to release of caged water to the bulk water and in our system can be facilitated by interactions between the peptide and the reactants.

The findings from the EE, HPLC, and CLSM analyses show a correlation between the recruitment of the reactant, the reaction rate, and the product localization (Fig. 6e). Specifically, low or moderate recruitment of AHC in WGI-0 and WGI-1 condensates at citrate buffer, respectively, results in low reaction rate with most of the product accumulates in the dilute phase. In contrast, increasing the reactant recruitment, as observed for WGI-0 and especially WGI-1 condensates in MES buffer, accelerates reaction rate and facilitates product accumulation inside the condensates.

Altogether, these results demonstrate that peptide hydrophobicity, as expressed by incorporation of a single aliphatic amino acid to the peptide sequence, affect the recruitment of a hydrophobic reactant, and in turn, reaction conversion, rate, and product localization. Thus, balancing microenvironment hydrophobicity by careful sequence design is critical for regulating reaction rate and conversion. This approach can be further expanded to increase the efficiency of reactions in water by replacing organic solvents with aliphatic microenvironments generated by biomolecular condensates.

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Competing interest statement

The authors declare no competing interests.

Methods

Materials and reagents

Peptides were custom synthesized and purified by GenScript (Hong Kong). 4-Ethynylanisole, Rhodamine B, t-BtOH and sodium sulfate (NaSO4), manufactured by ACROS organics; AHC by Apollo Scientific, copper sulfate ($CuSO_4$) by Fisher Bioreagents; Sodium ascorbate (NaAsc) by Glentham Life Sciences; Copper (ii) chloride dihydrate ($CuCl_2 \cdot 2H_2O$) by combi blocks; 7-Hydroxycoumarine (coumarin) by Aaron Chemical; acetonitrile (ACN) by biolab. Buffer salts: sodium citrate by Chem-Impex International, citric acid by Alfa Aesar and MES by Glentham Life Sciences.

Peptide solution preparation

All peptide condensates were prepared as follows: The desired peptide was dissolved in 20 mM citrate/MES buffer pH=6 at a final concentration of 5, 10, or 20 mM (final pH=5.5). In MES buffer, the pH was adjusted to 5.7 or 7.5. Condensates were formed following the addition of 10 mM $CuSO_4$.

Phase diagrams

Condensates were prepared by mixing a stock solution of peptide (final concentration of 2.5, 5, and 10 mM) dissolved in 20 mM citrate buffer pH 6 with 2% (v/v%) of $CuSO_4$ dissolved in ddw (solution B) to a final concentration of 2.5, 5 and 10 mM. Condensates were formed

immediately upon mixing. All measurements were performed at room temperature. turbidity was measured in triplicates by an optical density at λ=400 nm in a BioTek H1 synergy plate reader using 384-well plate. A bright field imaging was taken for samples at reported concentrations by fluorescence microscope (Olympus IX83), x40/0.95 NA Universal Plan Extended Apochromat objective. Images were collected and processed using CellSens Dimension software.

FRAP analysis

Fluorescence recovery after photobleaching (FRAP) experiments were performed using a Zeiss 900 LSM confocal microscope by tracking the fluorescent signal of 10 μM rhodamine B as payload**.** All the solutions were imaged on a slide coated by a solution of Pluronic F-127 surfactant, dissolved in ddw at a concentration of 10 mg/ml. Photobleaching was performed to a circular area with a diameter of 1.2 μm using 17 iterations of λex = 488 nm laser at 100% intensity, and subsequent recovery of the fluorescence at the bleached area was recorded and analyzed Zen Blue 3.2 software. Photobleaching correction and recovery time were calculated using OriginLab 9.95. Each experiment was normalized between 0% (bleach intensity) and 100% (intensity before bleaching). Then, the normalized data was fitted to the next equation to extract the fitting parameters: $y = a - bc^x$, when a is the recovery that was measured. $t_{1/2}$ values of WGI-0 and WGI-1 were calculated for each experiment using a,b,c extracted from the fitting, with the following equation: $t_{1/2} = \log_c\left(\frac{0.5*a}{b}\right)$. The final FRAP recovery curve is the average of recovery curves collected from n=4-8 separate condensates.

Product synthesis and purification

A mixture of 4-ethynylanisole (100 mg, 109 μL) and AHC (152.4 mg) in DMF (2 mL) was prepared. To this solution, $CuSO_4$ (188 mg) dissolved in 1 mL of ddw and NaAsc (223 mg)

dissolved in 1 mL of ddw were added sequentially. The reaction was stirred overnight under an argon atmosphere. Upon completion, the reaction mixture was quenched with ddw and extracted three times with ethyl acetate. The organic layer was concentrated under reduced pressure to afford the crude product. Yield: 83%. H^1 NMR (400 MHz, DMSO) δ 8.89 (s, 1H), 8.63 (s, 1H), 7.88 (d, 2H), 7.76 (d, 1H), 7.05 (d, 2H), 6.91 (d, 1H), 6.88 (s, 1H), 3.81 (s, 3H). LCMS: 336.31 $[M+H]^+$.

Product calibration curve

Product was dissolved at 1 mM stock solution in ACN. Then, diluted with ACN to 0.2, 0.4, 0.5, 0.6, 0.8 mM. all solutions were diluted once more with 5% t-BtOH (%v/v) in ddw (ratio 1:1), to compare the kinetics injection conditions. The product was monitored by HPLC, analytical reverse-phase high-performance liquid chromatography (RP-HPLC, Thermo Fisher), a Dionex SD Ultimate 3000 UHPLC standard system equipped with a diode-array detection (DAD) detector. Mobile phases were (A) H_2O (0.1% trifluoroacetic acid; TFA) and (B) MeCN (0.1% TFA), stationary phase was CS chromatography MultoHigh C18 column (250 mm × 4.6 mm, 5 μm particle size and 100 Å pore size, 5861272). Gradient: 35% to 40% solvent B in 24 minutes at flow rate of 1 mL/min. Area of the peak was analyzed at absorption of 214 nm, retention time: 20.3 minutes.

AHC calibration curve

AHC was dissolved in a stock solution of 20 mM in t-BtOH. From this solution, a serial dilution was made for the following stock solutions: 16, 12, 10, 8, 6, 4, 2 mM. From each stock solution, 5 μL were diluted in 95 μL ddw to final concentrations: 1, 0.8, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1 mM, then 100 μL ACN was added to compare to the kinetics conditions. AHC was monitored by HPLC, Gradient: 35% to 40% solvent B in 24 minutes at flow rate of 1 mL/min. Area of the peak was analyzed at absorption of 214 nm, retention time: 11.7 minutes.

Coumarin calibration curve

Coumarin was dissolved in a stock solution of 10 mM in tert-butanol. This solution was diluted in ddw and tert-butanol, to the following concentration:0.5, 0.4, 0.3, 0.2,0.1 mM, while tert butanol was kept at a final concentration of 5% v/v. this solution was further diluted with ACN (1:1). coumarin was monitored by HPLC, Gradient: 35% to 40% solvent B in 24 minutes at flow rate of 1 mL/min. Area of the peak was analyzed at absorption of 214 nm, retention time: 8 minutes.

Preparation of reactions in condensates

Peptide solutions were prepared as described above. AHC and 4-Ethynylanisole (both reactants were dissolved in 10 mM t-BtOH) were added to peptide solution at a final concentration of 0.5 mM. NaAsc (stock solution 1M in ddw) was added at a final concentration of 20 mM. Then, $CuSO_4$ (stock solution 1M in ddw) was added at a final concentration of 10 mM, following incubation of 30 minutes in argon. Condensates were formed immediately upon addition of $CuSO_4$.

Reaction kinetics HPLC analysis

For each condensate system, ACN was added at a final volume percentage of 50% v/v to quench the reaction at the desired time point (0, 5, 10, 15, 20, 40, 60, 80, 100 minutes). Product formation was monitored over time using HPLC. Gradient: 35% to 40% solvent B in 24 minutes at flow rate of 1 mL/min. Analysis was at λ=214 nm to monitor the reactants/product. Data points represent averages of at least three independent measurements. Statistical test used to compare the reactions rates was 2-sided t-test. All p-values data are presented in Tables S1-3 in the SI.

Encapsulation efficiency (EE) analysis

Peptides were dissolved at 10 mM in buffer as previously described, unless otherwise specified. Then the following solutions was added: 10 mM AHC/coumarin in t-BtOH (final concentration of 0.5 mM), ddw (2%, v/v%), and $CuSO_4$/$NaSO_4$/$CuCl_2$ dissolved in 1 M in ddw (final

concentration 10 mM). Droplets were formed when $CuSO_4/NaSO_4/CuCl_2$ was added. Then, the droplet solutions were centrifuge at 20℃, in 17K rcf for 30 minutes. 100 µL from the supernatant were diluted with 100 µL ACN. AHC/coumarin was monitored by HPLC, Gradient: 35% to 40% solvent B in 24 minutes at flow rate of 1 mL/min. Peak area was analyzed at λ=214 nm, AHC retention time: 11.7 minutes, coumarin retention time: 8 minutes. Concentration in supernatant was calculated using calibration curves. %EE was calculated with the following equation: $\left(\frac{C_i - C_{sup}}{C_i}\right) \times 100$. Values are based on 4 independent analyses for 10 mM and 20 mM WGI-1 in MES buffer pH 5.7, 6 independent analyses for 10 mM WGI-0 in MES pH 7.5 and all other values are based on 5 independent analyses for each system.

Confocal microscopy analysis of reaction kinetics

For each condensate system, condensates sample with reaction was prepared as described above. 35 µl of the reaction were transferred to a coated 96-well plate (coated with 100 µl of Pluronic F-127 surfactant). The solution was imaged using a Zeiss LSM 900 inverted confocal microscope, using an λ_{ex}= 405 nm laser, and collection emission range of λ_{em}= 410-700 nm. Images were taken over time, for 20 minutes. The collected images were taken using z-stack mode. Condensates of each sample were analyzed using the Zen blue 3.2 software (Zeiss) software to show the average fluorescence in condensates over time, and background fluorescence over time. 5 mM WGI-1 in citrate buffer: at 0 mins n=23, 10 mins n=53, 20 mins n=20; 5 mM WGI-0 in citrate buffer: at 0 mins n=30, 10 mins n=32, 20 mins n=34. 20 mM WGI-1 in MES buffer: at 0 mins n=30, 10 mins n=35, 20 mins n=25; 20 mM WG0-1 in MES buffer: at 0 mins n=22, 10 mins n=33, 20 mins n=21.

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